

A model-checking approach for verifying OSEK/VDX applications

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Abstract

OSEK/VDX applications are widely used in automotive embedded systems, where software components may directly affect vehicle operation and driver safety. Errors in such applications can therefore lead not only to incorrect program behaviour, but also to reliability and safety risks. However, these applications are difficult to verify using testing alone, because testing may miss important execution cases, and errors are often hard to reproduce. Model checking is a suitable verification technique for this setting because it can explore possible behaviours and provide counterexamples when a property is violated.

OSEK/VDX applications consist of multiple tasks whose execution order is determined by the OSEK/VDX scheduler. During execution, tasks invoke standardized application interfaces (APIs), such as `ActivateTask`, `TerminateTask`, `WaitEvent`, and `SetEvent`, to change task states, synchronization events, shared resources, and the ready queue. Therefore, a checking model for OSEK/VDX applications should reflect the behaviours of the scheduler instead of treating these APIs as ordinary function calls only.

Existing model checking methods can exhaustively check concurrent software, but they may become imprecise when the OSEK/VDX scheduler is not represented explicitly. If unnecessary task interleavings are explored during verification, the checker may report spurious bugs. This increases the effort required from developers, because they must determine whether a reported bug is a real error or only an artifact of an inaccurate checking model.

This report presents a checking-model approach for OSEK/VDX applications. The approach has two main parts. First, it represents the effects of OSEK/VDX APIs on task states, events, shared resources, the ready queue, and task execution order. Second, it uses a manifest to record selected program locations where interrupt service routine (ISR) calls should be inserted. The final checking model contains guarded ISR calls only at these selected locations, rather than inserting ISR calls broadly across the program.

The experimental results show that the proposed approach improves the verification process for OSEK/VDX applications. It builds a more accurate checking model, reduces unnecessary interleavings, removes spurious bugs reported by the baseline checking process, and makes the checking process clearer.

1 Introduction

OSEK/VDX [1, 2] is an automotive operating-system standard introduced in 1994 by European automobile manufacturers, suppliers, and the University of Karlsruhe. It was created to manage the increasing amount of software in vehicles while improving product quality and reliability. The standard has been adopted by major car manufacturers such as BMW, Opel, and Toyota, and is used to support vehicle applications such as cruise control and temperature control. Because these applications directly affect vehicle operation and driver safety, ensuring their reliability is essential. However, as automotive software becomes more complex, verifying the correctness and reliability of OSEK/VDX applications remains a significant challenge.

1.1 OSEK/VDX Applications and Model Checking

An application developed on OSEK/VDX OS consists of multiple tasks. To concurrently execute multiple tasks on one CPU, OSEK/VDX scheduler uses *static priority scheduling policy* with non-preemptive strategy and full-preemptive strategy to determine the running task. In particular, Figure 1 shows that a ready queue in scheduler is used to lay out the execution order of tasks. OSEK/VDX OS can process two types of tasks, basic task and extended task. The states of a basic task consist of *running state*, *ready state* and *suspended state*. Compared with the basic task, the extended task can hold synchronization events and has a unique state called *waiting state*.

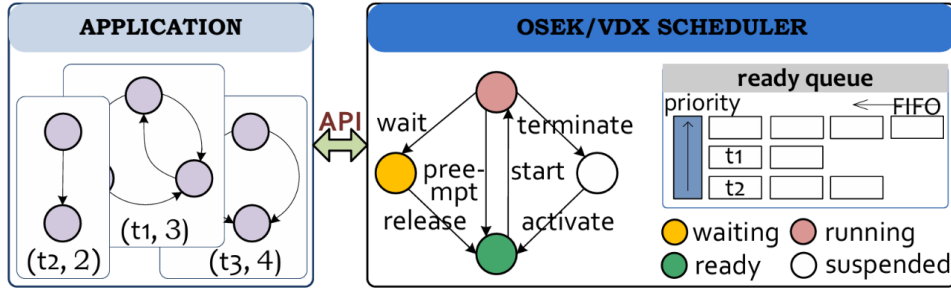


Figure 1: OSEK/VDX scheduler

OSEK/VDX OS also provides standardized application interfaces (APIs), such as **ActivateTask**, **TerminateTask**, **SetEvent**, and **WaitEvent**, through which tasks can dynamically change task states, synchronization events, and shared resources.

To ensure the reliability of OSEK/VDX applications, testing is an important checking technique used in industry, but it is difficult to exhaustively check OSEK/VDX applications using testing technique, especially because of incomplete test-cases. In addition, it is hard to reproduce an error trace when an error is detected by testing, due to the concurrency of OSEK/VDX applications.

Model checking [3, 4], an efficient and exhaustive verification technique for concurrent software, can be applied to verify OSEK/VDX applications instead of testing technique. Particularly, it will return a counterexample when detecting an error. Based on the returned counterexample, developers can easily find the error trace and reproduce the erroneous behaviours.

1.2 Shortcomings of Existing Model Checking for OSEK/VDX Applications

Existing model checking methods [5, 6, 7], such as the ANSI-C multi-threaded software and SystemC multi-threaded software, have been applied to verify concurrent software. In these methods, the behaviours of scheduler are taken into account in the verification process. However, these prominent model checking methods are still unsuitable to accurately verify OSEK/VDX

applications because of different scheduling policy. As shown in Figure 2, in the existing methods, the running thread within the concurrent software is arbitrarily determined by scheduler (non-deterministic scheduler is used to dispatch threads), but in OSEK/VDX applications which task is to be run is explicitly determined by OSEK/VDX scheduler, a deterministic scheduler used to dispatch tasks.

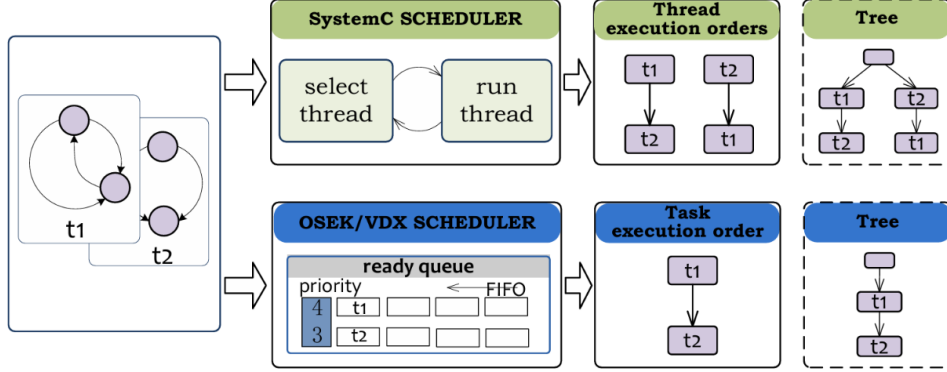


Figure 2: The difference of schedulers

If such methods are applied directly to OSEK/VDX applications, the verification process becomes too imprecise because the behaviours of OSEK/VDX scheduler are not taken into account. As a result, a large number of unnecessary interleavings of tasks may be checked in the verification stage, which can lead to spurious bugs and increases the checking costs in terms of judging time and memory of checking devices.

1.3 Research Purpose & Contribution

The purpose of this report is to make model checking more accurate for OSEK/VDX applications. In order to achieve this purpose, we describe and develop an approach to construct a checking model that reflects their execution semantics and avoid checking ISR occurrences at program locations that do not affect the checking results.

In our approach, we introduce how the checking model is constructed before the final verification stage. During checking-model construction, a recognized OSEK/VDX API call is not treated only as an ordinary C function call. Instead, it updates the representation of the current execution state, including the running task, the ready queue, the waiting tasks, and the stored event state of each relevant task. After processing a recognized API call, the checking model updates the current scheduling state. From this updated state, the deterministic scheduler determines what is checked next: execution may remain in the current task, switch immediately to a ready task, return to the interrupted context that preceded the current task, or finish the current task and start another ready task. In addition, some recognized API calls write the stored event state back to program variables, so later branch conditions and assertions can depend on that state. As a result, the checking model is constructed from these state updates and the resulting dispatch decisions, rather than from the ordinary local control flow of the current task alone.

To avoid checking unnecessary interleavings, our approach also constructs a manifest, which is an intermediate description used in checking-model construction: for each ISR, the manifest records the shared or global state that the function can write, and the source locations where that function needs to be represented in the checking model. This manifest is built by identifying the ISR or interleaving functions under consideration, summarizing the shared or global state that each of them can modify, and then scanning the remaining program to find the program points where reads, writes, branch conditions, or scheduling effects can be changed by that state.

The final checking model then introduces guarded ISR behaviour only at the source locations recorded in the manifest, so that the verification still reflects the task execution order determined by the deterministic scheduler while avoiding unnecessary interleavings and reducing spurious bugs.

The report makes three contributions:

- It describes a checking approach for OSEK/VDX applications whose task execution order is determined by a deterministic scheduler, task states, synchronization events, and a ready queue.
- It introduces a manifest-guided method for selecting ISR insertion locations and building a checking model with ISR calls only at the program locations recorded in the manifest.
- It presents benchmark results showing that the proposed approach reduces unnecessary interleavings and removes spurious bugs reported by the baseline checking process.

2 Background

2.1 OSEK/VDX OS

A general OSEK/VDX OS consists of a scheduler module, synchronization event process module, shared resource process module, alarm process module and interruption process module. Based on these system modules, OSEK/VDX OS supports a standardized application interfaces (APIs) for user to develop customized applications. In our research, we focus on the applications that communicate with scheduler module, synchronization event process module and resource process module. The structure of OSEK/VDX OS with an application is shown in Figure 3.

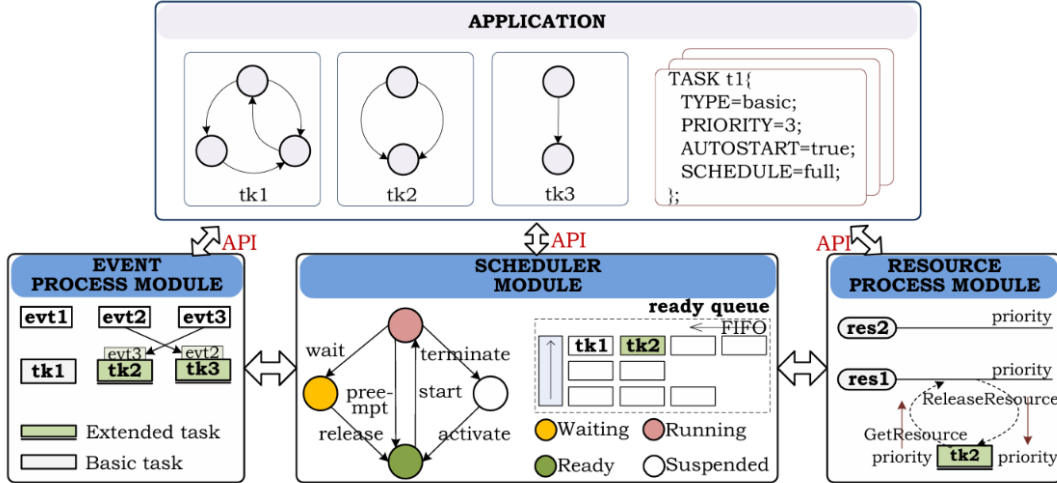


Figure 3: The structure of OSEK/VDX OS with an application

2.1.1 Scheduler Module

OSEK/VDX OS can process two types of tasks, basic task and extended task. The states of a basic task consist of *running state*, *suspended state*, and *ready state*. Compared with the basic task, the extended task can hold synchronization events and has a unique state called waiting state. In the scheduling process, the *static priority scheduling policy* with non-preemptive and full-preemptive strategies is adopted by the scheduler to conduct the executions of tasks. Particularly, in OSEK/VDX OS, scheduler manages a ready queue to lay out the execution order

of tasks. Based on the ready queue, the running task is explicitly determined by OSEK/VDX scheduler.

In addition, scheduler module provides several APIs for applications, such as *TerminateTask*, *ActivateTask* and *ChainTask*. Tasks within the application can invoke these APIs to dynamically change the states of tasks, and the changed states will affect the scheduling of tasks, e.g., the context switch of tasks will happen when a suspended task is activated by running task using API *ActivateTask* or *ChainTask*. For instance, as shown in Figure 4, there are two tasks $t1$ and $t2$, and the priority of task $t1$ is lesser than $t2$. Currently, task $t1$ is running task. When the API *ActivateTask*($t2$) is invoked by $t1$, scheduler will move task $t2$ from suspended state to ready state. At this moment, the context switch of tasks will happen, because the priority of task $t2$ is higher than $t1$.

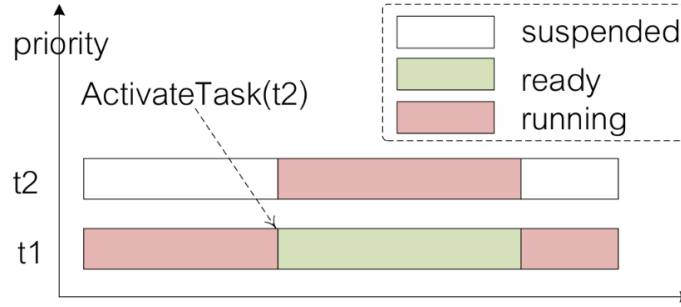


Figure 4: The context switch of tasks and API *ActivateTask*

2.1.2 Event Process Module

In the synchronization event process module, OSEK/VDX OS provides a synchronization mechanism for implementing synchronous executions between tasks. Particularly, only extended tasks can hold a definite number of events, and events are the criteria for the switching of task states from *running state* to *waiting state* or from *waiting state* to *ready state*. There are three APIs *SetEvent*, *WaitEvent* and *ClearEvent* that can be responded by event process module, and tasks within application can invoke these APIs to implement the synchronous behaviours. E.g., as shown in Figure 5, when the running task $t1$ waits for the event $evt1$ using API *WaitEvent*($evt1$), task $t1$ cannot continue until the event $evt1$ is set by other tasks (basic tasks or extended tasks) using API *SetEvent*($t1, evt1$).

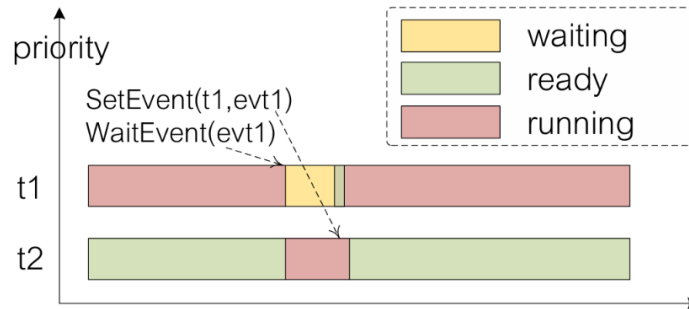


Figure 5: The synchronous behaviours between tasks

2.1.3 Resource Process Module

OSEK/VDX OS adopts the *Priority Ceiling Protocol* [8] to coordinate the behaviors of task accessing shared resources in the resource process module. The resource process module supports

two APIs (*GetResource* and *ReleaseResource*) that can be invoked by tasks to access shared resources. E.g., as shown in Figure 6, if the API *GetResource(res1)* is invoked by running task *t1*, and the priority of the task is lower than the ceiling priority of the resource *res1*, the priority of the task will be raised to the ceiling priority of the resource *res1*, and the priority of the task will be reset to the priority before requiring the resource *res1* when *ReleaseResource(res1)* is invoked by the task. Note that, the ceiling priority of a shared resource is lower than the lowest priority of all tasks that do not access the resource, and it is higher than the priorities of all tasks that access the resource.

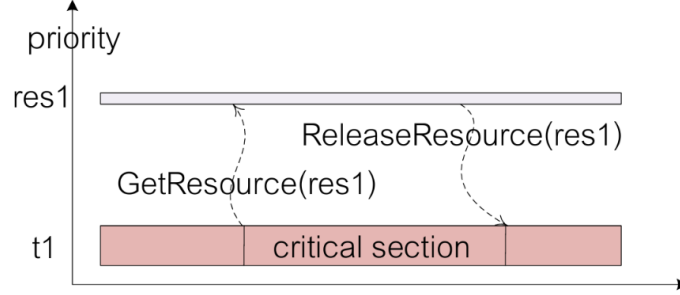


Figure 6: The priority ceiling protocol

2.2 OSEK/VDX Application and Execution Characteristics

2.2.1 OSEK/VDX Application

An application developed based on OSEK/VDX OS consists of two files, one is the source file, and the other is the configuration file. The source file which can be developed by C language is used to present the concrete behaviors of the application. The configuration file is used to define tasks, synchronization events and shared resources. In the configuration file, the attribute **AUTOSTART** is used to set the initial state of tasks. If the attribute **AUTOSTART** of a task is set to be **TRUE**, the task starts from ready state in the initial state (it will be inserted into ready queue according to the priority of the task). Otherwise, the task starts from suspended state. The attribute **SCHEDULE** is used to indicate the scheduling type. If the attribute **SCHEDULE** of a task is set to be **FULL**, the task can be preempted by higher priority tasks. Otherwise, the task cannot be preempted by higher priority tasks. A simple OSEK/VDX application without synchronization events and shared resources is shown in Figure 7. As to clearly comprehend the execution characteristics of OSEK/VDX applications, an example is symbolically executed in this part. In the application shown in Figure 7, only the attribute **AUTOSTART** of **t1** is set to be **TRUE**. Thus, **t1** will be firstly moved to running state by scheduler and then executed in the initial state.

Source file (.cpp)	Configuration file (.oil)
<pre> int a, b, c, h; TASK t1(){ if(b >= a){ a++; ActivateTask(t2); } else{ b++; ActivateTask(t3); } ActivateTask(t4); TerminateTask(); } TASK t2(){ if(h > 0) h--; else h++; TerminateTask(); } TASK t3(){ c=b; c=(c+b)/b; TerminateTask(); } TASK t4(){ assert(a > b); TerminateTask(); } </pre>	<pre> TASK t1{ TYPE=BASIC; SCHEDULE=FULL; PRIORITY=5; AUTOSTART=TRUE; }; TASK t2 { TYPE=BASIC; SCHEDULE=NON; PRIORITY=6; AUTOSTART=FALSE; }; TASK t3 { TYPE=BASIC; SCHEDULE=NON; PRIORITY=4; AUTOSTART=FALSE; }; TASK t4 { TYPE=BASIC; SCHEDULE=NON; PRIORITY=4; AUTOSTART=FALSE; }; </pre>

Figure 7: The structure of OSEK/VDX OS with an application

As shown in Figure 8, when the API *ActivateTask(t2)* is invoked by *t1* (the branch“(b >= a)” is chosen to symbolically execute), scheduler will be loaded to respond to the API (task *t2* is moved from suspended state to ready state). At this moment, the running task *t1* will be preempted by *t2* since the priority of *t2* is higher than *t1* and the attribute *SCHEDULE* of *t1* is set to be *FULL* (the context switch happens after the API *ActivateTask(t2)*). Currently, task *t2* gets run-unit to run, and goes to suspended state when the API *TerminateTask()* is invoked (*TerminateTask()* is used to terminate the executions of a task, and the terminated task will be moved from running state to suspended state by scheduler. If the running task is terminated, scheduler then dispatches the head task in the ready queue to run when ready queue is not empty). When task *t2* terminates itself using API *TerminateTask()*, scheduler will dispatch task *t1* to running state. Then, *t1* continues its executions from the preempted point, and task *t4* is activated by *t1* using API *ActivateTask(t4)* (the context switch will not happen, since the priority of task *t4* is less than *t1*). Finally, task *t4* will be executed when the running task *t1* terminates itself using API *TerminateTask()*. Thus, according to the conducted symbolic executions, we can get a task execution sequence *t1,t2,t1,t4* in the execution path π_1 , and π_2 , shown in Figure 8. However, if the API *ActivateTask(t3)* is invoked by task *t1* (the “branch !(b >= a)” is chosen to symbolically execute), we will get a different task execution sequence *t1,t3,t4* in the execution path π_3 .

2.3 Model Checking

Concurrent software usually consists of multiple threads. To concurrently execute multiple threads on one CPU, a scheduler is used to dispatch threads. There have been many concurrent software standards such as SystemC standard, POSIX standard and ANSI C standard that have been widely used in the industry of software. In these concurrent software standards, which thread within software is to be run is arbitrarily determined by scheduler. In other words, the running thread is stochastic when the multi-threaded software runs on the machine. If testing technique is used to check such software, it is difficult to reproduce an error track because of the non-deterministic scheduling. In contrast to testing technique, model checking can solve the problem based on the reported counterexample.

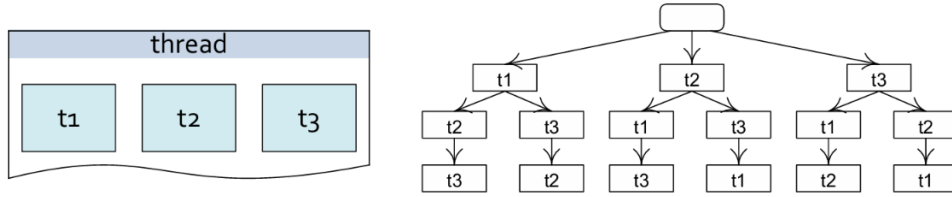


Figure 9: The model for a SystemC multi-threaded software

The key idea of model checking is: firstly construct a model to describe the executions of a target system, and then search the constructed model with given property. To verify concurrent software, a tree as a model is usually constructed for demonstrating all of the possible interleavings of threads. E.g., as shown in Figure 9, there is a SystemC multithreaded software, and the multi-threaded software consists of three threads. In model checking, a tree shown in the right side of Figure 9 as a model will be constructed to exhaustively describe all of the possible executions of the multi-threaded software. When the model of concurrent software is constructed, there existing many checking techniques that can be applied to search the constructed model for checking whether the constructed model satisfies the given property or not, such as explicit state checking technique [9], binary decision diagrams (BDD) technique [10, 11], symbolic checking technique [12] and bounded model checking technique [13].

Currently, based on the model checking technique, there are many model checkers that have been established based on the different model checking techniques, and these established model checkers have been successfully applied to verify multi-threaded software that conforms to different concurrent software standards. E.g., Bell Labs developed a model checker named **spin** [5] based on the explicit state model checking technique. For the ANSI-C standard, a bounded model checking based model checker named **esbmc** [14] has been implemented in the university of Southampton. In addition, in the group of Cimatti, a model checker named **kratos** [15] has been established for SystemC multithreaded software. There are also several famous model checkers, such as symbolic model checker **SMV** [16], bounded model checker **cbmc** [17], etc.

2.4 Bounded Model Checking

The original motivation of bounded model checking was to leverage the success of satisfiability (SAT) [18, 19] in solving Boolean formulas to model checking. Compared to explicit state model checking technique and BDD-based model checking technique [10, 11], it offers the advantage of handling the verification of large state spaces. The main idea of BMC is to avoid the full state space generation and look for witnesses of an existential specification on the suitable subsets of the full model under the given bound. Once a sub-model is selected, the considered sub-model will be translated into propositional formulae. Then, the propositional formulae and the negative form of given specification (given property) will be solved by a specialized SAT solver. If the verification result is true, it means that the sub-model does not satisfy the given specification, a

Aspect	Verification flow	Proposed Approach
Task execution order	Follows the ordinary C program model unless OSEK/VDX scheduling is encoded separately.	Follows task states, the ready queue, and the deterministic scheduler inside the checking model.
OSEK/VDX APIs	May be treated as ordinary function calls if no OSEK/VDX-specific model is provided.	Are handled as operations that can change task states, synchronization events, shared resources, and the ready queue.
Interrupt behaviour	May represent ISR occurrences broadly at many program locations.	ISR calls are inserted only at selected locations recorded in the manifest.
Interleavings	May include unnecessary interleavings that increase the checking cost and may lead to spurious bugs.	Reduces unnecessary interleavings before the final verification step.
Checking model	Represents the program without explicitly combining OSEK/VDX scheduler behaviours and selected ISR locations.	Represents the OSEK/VDX scheduler, changes caused by APIs invoked from tasks, and selected ISR insertion points before final verification.

Table 1: Conceptual comparison between verification flow and the proposed approach.

counterexample will be report by the SAT solver. Otherwise, a larger sub-model will be selected and the whole procedure will be run again until the whole model is verified.

3 Methodology

3.1 Overview of the Proposed Approach

The proposed approach focuses on how the checking model is constructed before final verification. It does not depend on replacing a particular verification tool. Instead, the main goal is to build a more accurate checking model for OSEK/VDX applications, where task execution order is determined by the deterministic scheduler, the ready queue, task states, and the APIs invoked from tasks.

The approach has two main parts. The first part is to represent the behaviours of the OSEK/VDX scheduler in the checking model. During verification, the model must capture how OSEK/VDX APIs can change task states, synchronization events, shared resources, and the ready queue. This allows the next running task to be determined according to the OSEK/VDX scheduler rather than ordinary C control flow alone.

The second part is manifest-guided interrupt service routine (ISR) insertion. In this report, an ISR is a function that may interrupt the normal execution of tasks. In practice, an ISR may occur at many points during task execution. However, the checking model does not need to represent every possible ISR occurrence separately. The proposed approach focuses on the program locations where an ISR call can affect the checked execution, such as by changing task states, event state, shared variables, the ready queue, or the checked assertion. Other locations are skipped because they only add unnecessary interleavings and increase the checking cost. To avoid this, the proposed approach first computes a manifest, which records the selected program locations where ISR calls should be inserted. The final checking model is then built with guarded ISR calls only at the locations recorded in the manifest. Table 1 summarizes the difference between verification flow and the proposed approach.

The key point is that the proposed approach does not replace the underlying verification tool. Instead, it changes the checking model given to the final verification step. Rather than checking a model that represents scheduling and ISR occurrences too broadly, the approach

constructs a checking model that better reflects the executions of OSEK/VDX applications.

The remainder of this section explains the approach in three steps. Section 3.2 describes how OSEK/VDX scheduler behaviours and APIs invoked from tasks are represented. Section 3.3 explains how the manifest is used to reduce unnecessary ISR insertion points. Section 3.4 explains how the checking model that takes the OSEK/VDX scheduler into account and the manifest-guided insertion step are combined before CBMC performs final verification.

To keep the discussion concrete, the remainder of this section uses one synthetic running example. *Task1* is a higher-priority extended task that waits for an event. *Task2* is a lower-priority task that can run while *Task1* is in waiting state. *ISR1* updates shared state and sets the event awaited by *Task1*. The example is intentionally small, but it is large enough to show the three main decisions in the methodology: how task execution order is determined, how unnecessary interleavings are reduced, and how the final checking model is built.

3.2 Taking the OSEK/VDX Scheduler into Account

The first methodological step is to represent the behaviour of the OSEK/VDX scheduler inside the checking model. In an OSEK/VDX application, the next running task is not chosen by an unstructured interleaving of threads. It is chosen by a deterministic scheduler that depends on task states, priorities, events, and the ready queue. If the checking model ignores those elements, the resulting task execution order no longer matches the execution order of the application.

Listing 1 shows the running example used throughout this section. The example combines waiting, event signaling, shared-state update, and an local computation so that each later step can be explained with reference to the same program.

Listing 1: Pseudo Code

```
int shared_ready = 0;
int shared_value = 0;

TASK(Task1)
{
    EventMaskType observed = 0;
    WaitEvent(EVENT_SHARED_READY);
    GetEvent(TASK_1_ID, &observed);
    if ((observed & EVENT_SHARED_READY) != 0u && shared_ready) {
        int sample = shared_value;
        assert(sample >= 0);
    }
    ClearEvent(EVENT_SHARED_READY);
    TerminateTask();
}

TASK(Task2)
{
    int checksum = 0;
    checksum = checksum + 1;
    checksum = checksum * 2;
    TerminateTask();
}

ISR(ISR1)
{
    shared_value = 7;
    shared_ready = 1;
    SetEvent(TASK_1_ID, EVENT_SHARED_READY);
}
```

Table 2 summarizes the intended execution context for this example. *Task1* starts as the running task, *Task2* is already ready with a lower priority, and *ISR1* can make *Task1* ready again by setting the event that *Task1* is waiting for.

Entity	Priority / type	Initial status and role
<i>Task1</i>	Higher-priority extended task	Running at the beginning of the path; waits for <code>EVENT_SHARED_READY</code> ; later reads the event state and the shared value.
<i>Task2</i>	Lower-priority task	Ready while <i>Task1</i> is running; may execute while <i>Task1</i> is in waiting state; performs only local computation in this example.
<i>ISR1</i>	Interrupt service routine	Writes <code>shared_ready</code> and <code>shared_value</code> , then sets the event awaited by <i>Task1</i> .

Table 2: Configuration summary for the running example.

The central point is that the standardized APIs in Listing 1 cannot be treated as ordinary function effects. When *Task1* executes `WaitEvent`, the important effect is not only that a C function is called. The important effect is that *Task1* may move from running state to waiting state. Once this happens, the scheduler may select *Task2* as the next running task because *Task2* is already in ready state.

Later, when *ISR1* executes `SetEvent`, the event state changes and *Task1* may move from waiting state back to ready state. Because *Task1* has higher priority than *Task2*, the deterministic scheduler may dispatch *Task1* before *Task2* continues. Therefore, the checking model must represent changes to task states, events, and the ready queue, not only ordinary control-flow changes inside the current C function.

Table 3 is the output of this first methodological step. It shows how task states, the ready queue, and the event state change when the same source program is checked under the OSEK/VDX model.

Step	Current context	Task states	Ready queue	Event state	Scheduler action
0	Start of path	<i>Task1</i> : running; <i>Task2</i> : ready	<i>Task2</i>	Clear	Continue in <i>Task1</i>
1	<i>Task1</i> executes <code>WaitEvent</code> while the event is clear	<i>Task1</i> : waiting; <i>Task2</i> : ready	<i>Task2</i>	Clear	Dispatch <i>Task2</i>
2	<i>Task2</i> executes local computation	<i>Task1</i> : waiting; <i>Task2</i> : running	Empty	Clear	Continue in <i>Task2</i> unless <i>ISR1</i> occurs
3	<i>ISR1</i> updates shared state and executes <code>SetEvent</code>	<i>Task1</i> : ready; <i>Task2</i> : running before dispatch	<i>Task1</i>	Set	Dispatch <i>Task1</i> because it has higher priority
4	<i>Task1</i> continues, observes the event, and reads the shared value	<i>Task1</i> : running; <i>Task2</i> : ready	<i>Task2</i>	Set, then cleared	Continue in <i>Task1</i>

Table 3: State evolution for the running example.

Figure 10 gives a high-level view of how scheduling information is produced in the checking

model. When a task invokes an OSEK/VDX API, the API may change task states or events. These changes update the ready queue, and the deterministic scheduler uses the ready queue to decide which task should execute next. Therefore, the checking model must represent this cycle explicitly instead of treating API calls as ordinary function calls.

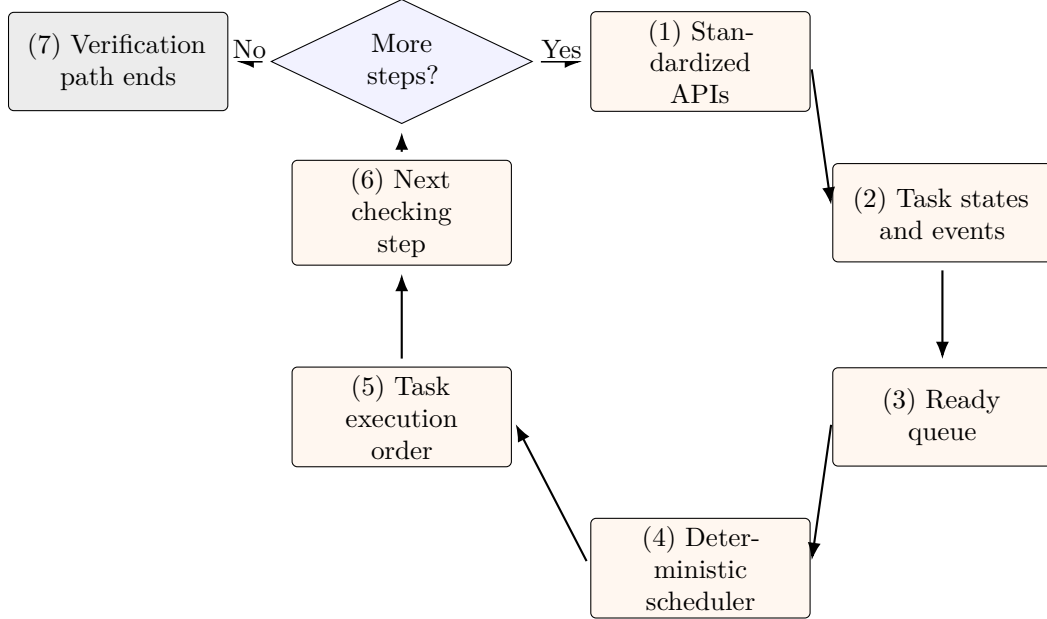


Figure 10: Conceptual cycle from standardized application interfaces (APIs) to task states, the ready queue, the deterministic scheduler, and the next checking step.

Read clockwise, some OSEK/VDX APIs change task states or event state. For example, *WaitEvent* may move an extended task from running state to waiting state when the awaited event is not set, while *SetEvent* may make a waiting task ready again by setting the awaited event. Other APIs, such as *GetEvent*, read event information that is already stored in the checking model. After each relevant API action, the checking model updates the task states, the event state, and the ready queue. The deterministic scheduler then uses this information to determine the next running task. That task execution order determines the next checking step, which may later reach another API call or another state update. If the verification path reaches its end, the loop stops at the termination branch.

In the running example, this means that the verification path should not simply continue with the next statement in the C code after *WaitEvent* when the awaited event is clear. Instead, the checking model should represent that *Task1* moves to waiting state, that *Task2* may run while *Task1* is waiting, and that *ISR1* can set the event and make *Task1* ready again. In this way, the task execution order in the verification result can reflect the execution order determined by the OSEK/VDX scheduler.

3.3 Reducing Unnecessary Interleavings

After the checking model represents the OSEK/VDX scheduler, the next step is to control where interrupt service routine (ISR) calls are inserted. If ISR calls are inserted at too many program locations, the checking model may include many interleavings that do not affect the property being checked. These unnecessary interleavings can increase the checking costs in terms of states, time, and memory, and may also lead to spurious bugs.

Therefore, the proposed approach first identifies the program locations where ISR calls are relevant to the checked execution. These selected locations are recorded in a manifest. The later

rewriting step uses this manifest to insert ISR calls only at those selected locations, instead of adding them broadly across the program.

The key question is therefore not “where could an ISR call be inserted in general?”, The key question is “where can an ISR call change the checking result?”. In the running example, *ISR1* is relevant when its updates to `shared_ready` and `shared_value`, or its call to `SetEvent`, can be observed later by *Task1* or by the scheduler.

If inserting *ISR1* at a program location cannot change a task state, event state, ready queue, or assertion result, then that insertion only adds an unnecessary interleaving to the checking model.

In the running example, *ISR1* has three visible effects: it writes `shared_value`, sets `shared_ready`, and calls `SetEvent(TASK_1_ID, EVENT_SHARED_READY)`. These effects matter when they can be used later by *Task1* or by the scheduler. For example, the call to `SetEvent` can move *Task1* from waiting state back to ready state, and the writes to `shared_ready` and `shared_value` can affect the later `if` statement and assertion in *Task1*.

By contrast, the checksum statements in *Task2* only update the local variable `checksum`. They do not read `shared_ready` or `shared_value`, they do not inspect the event state, and they do not invoke an OSEK/VDX API that changes task states or the ready queue. Therefore, these local statements do not create additional points.

Table 4 shows gives a readable summary of the selected program locations: where an *ISR1* call should be inserted, and where a separate insertion would only add an unnecessary interleaving.

Candidate program location	Can <i>ISR1</i> change results here?	Decision	Reason
Immediately before <code>WaitEvent(EVENT_SHARED_READY)</code>	Yes	Insert	<i>ISR1</i> can set <code>EVENT_SHARED_READY</code> . This can change whether <i>Task1</i> continues or moves to waiting state.
Immediately before <code>GetEvent(TASK_1_ID, &observed)</code>	Yes	Insert	The value stored in <code>observed</code> depends on whether <code>EVENT_SHARED_READY</code> has already been set.
Immediately before the <code>if</code> statement in <i>Task1</i>	Yes	Insert	The branch checks both <code>observed</code> and <code>shared_ready</code> , which can be changed by <i>ISR1</i> .
Immediately before the read of <code>shared_value</code>	Yes	Insert	The value read by <i>Task1</i> is written by <i>ISR1</i> .
Immediately before <code>checksum = checksum + 1</code> in <i>Task2</i>	No	Skip	This statement only updates the local variable <code>checksum</code> . It does not read the event state, <code>shared_ready</code> , or <code>shared_value</code> .
Immediately before <code>checksum = checksum * 2</code> in <i>Task2</i>	No	Skip	This statement is also local to <i>Task2</i> . A separate insertion here would add another interleaving without adding a new observation point in this example.

Table 4: Summary of selected ISR insertion locations in the running example.

The point of Table 4 is not to predict exactly when *ISR1* will occur. The point is to decide where inserting an *ISR1* call is useful for the checking model. The selected locations are the places where *ISR1* can change what *Task1* or the scheduler observes, such as the event state, `shared_ready`, `shared_value`, or the next running task.

The skipped locations are different. In this example, the checksum statements in *Task2*

only update the local variable `checksum`. They do not read the event state, they do not read `shared_ready` or `shared_value`, and they do not change task states or the ready queue. Skipping separate ISR insertions at those locations therefore reduces unnecessary interleavings without removing behaviour that can affect the checked path.

3.4 Building the Final Checking Model

The proposed approach combines the previous two steps to build the final checking model. First, it selects the program locations where an *ISR1* call can change what *Task1* or the scheduler observes. These locations are recorded in the manifest. Second, it uses the manifest to build a checking model in which ISR calls appear only at the selected locations.

This separation makes the method easier to control. The selection step answers the question: “where can an ISR call affect the event state, the ready queue, the shared variables, or the assertion checked?” The model-building step answers a different question: how should the final checking model represent only those selected ISR calls?

As a result, the final checking model does not insert ISR calls at every possible program location. It keeps the locations that can affect the checked path and skips locations that only add unnecessary interleavings. This reduces the checking cost while preserving the behaviours needed for the running example.

For the running example, the selected locations can also be summarized in the compact form shown in Listing 2. This summary connects the selection step with the model-building step: it records where *ISR1* should be inserted, without showing low-level implementation details.

Listing 2: Readable summary of the selected ISR insertion locations for the running example.

Selected ISR insertion locations:
1. Before <code>WaitEvent(EVENT_SHARED_READY)</code>
2. Before <code>GetEvent(TASK_1_ID, &observed)</code>
3. Before the if statement that checks <code>observed</code> and <code>shared_ready</code>
4. Before the read of <code>shared_value</code>
Skipped locations: The local checksum updates inside <code>Task2</code>

Figure 11 shows how the proposed approach builds the final checking model before bounded model checking. The upper lane is the verification flow. The lower lane adds four stages before the final verification step.

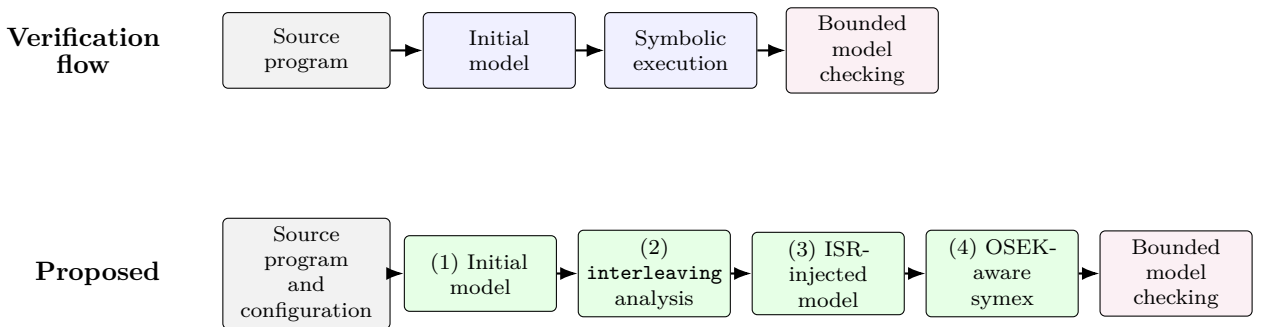


Figure 11: Baseline and improved checking-model construction.

In the verification flow, the source program and the OSEK/VDX configuration are translated into an initial checking model, then we perform symbolic execution and model checking. This path is useful for ordinary C programs, but it does not by itself explain how OSEK/VDX APIs should change task states, events, the ready queue, and task execution order.

The proposed flow adds four stages. First, the extended `goto-cc` stage builds the initial program representation from the source program and the OSEK/VDX configuration. Second, `interleaving-analysis` selects the program locations where ISR calls may affect the checking model. Third, `aib` uses those selected locations to build a restricted checking model, so ISR calls are added only where they are needed. Fourth, `goto-symex` with the `os-api` layer symbolically executes the model while handling OSEK/VDX APIs as scheduler operations rather than ordinary C function calls.

Listing 3 shows the output of the model-building step for the running example. The inserted calls are part of the checking model, not part of the original application program. Each inserted call is guarded by `nondet_bool()`, so model checking may choose whether *ISR1* occurs at that selected location. No ISR call is inserted inside the local checksum updates of *Task2*, because those updates only change the local variable `checksum` and were skipped in Table 4.

Listing 3: Illustrative checking model with selected ISR calls for the running example.

```

TASK(Task1)
{
    EventMaskType observed = 0;

    if (nondet_bool()) ISR1();
    WaitEvent(EVENT_SHARED_READY);

    if (nondet_bool()) ISR1();
    GetEvent(TASK_1_ID, &observed);

    if (nondet_bool()) ISR1();
    if ((observed & EVENT_SHARED_READY) != 0u && shared_ready) {
        if (nondet_bool()) ISR1();
        int sample = shared_value;
        assert(sample >= 0);
    }

    ClearEvent(EVENT_SHARED_READY);
    TerminateTask();
}

TASK(Task2)
{
    int checksum = 0;
    checksum = checksum + 1;
    checksum = checksum * 2;
    TerminateTask();
}

```

The role of Listing 3 is simple. The final checking model still allows *ISR1* to occur, but only at the selected locations. These locations are kept because *ISR1* can change the event state, the shared variables `shared_ready` and `shared_value`, or the next running task chosen by the scheduler.

By contrast, no ISR call is inserted inside the local checksum updates of *Task2*. Those statements do not read the event state or the shared variables, and they do not change task states or the ready queue. Skipping them reduces unnecessary interleavings while preserving the OSEK/VDX scheduler behaviours needed for the checked execution.

4 Implementation and Experiments

4.1 Tool Implementation

CBMC is a bounded model checker for C programs [13, 17, 20]. Its verification flow includes front-end translation, symbolic execution, and solver-based checking. This makes CBMC a suitable implementation target for this work for two reasons.

First, the approach in this report and the benchmark cases used in the evaluation are written in C. Therefore, CBMC can be used directly as the back-end checker after the checking model has been constructed.

Second, CBMC provides clear implementation points for the proposed methodology. The initial program representation can be built around `goto-cc`, selected ISR insertion locations can be computed before final verification, and OSEK/VDX API semantics can be handled during symbolic execution. Thus, the implementation does not replace the underlying bounded model checking engine. It changes the checking model and the symbolic execution behaviour so that the final verification step receives a model that better reflects the executions of OSEK/VDX applications.

The CBMC verification flow used in this report is shown in Figure 12. Source code is translated by the front-end into an internal program representation, that representation is symbolically executed to construct a solver formula, and the solver determines whether the required property holds under the chosen bound.

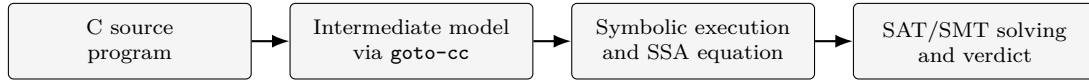


Figure 12: CBMC architecture used as the implementation target for the proposed approach.

This flow is a strong starting point for C verification, but it is not sufficient by itself for OSEK/VDX applications. The limitation is not symbolic execution itself. The limitation is that the checking model must also represent OSEK/VDX scheduling behaviours and must avoid checking ISR calls at program locations that do not affect the execution path.

The current implementation realizes the methodology inside CBMC by extending the flow in four places. The `goto-cc` extension builds the initial checking model while preserving the OSEK/VDX project context. The `interleaving-analysis` stage produces a manifest of selected ISR insertion locations. The `aib` component uses this manifest to build the checking model with guarded ISR calls only at the selected locations. Finally, `goto-symex` is extended with the `os-api` layer so that recognized OSEK/VDX APIs update task states, events, the ready queue, and the next running task during symbolic execution.

Table 5 maps these implementation choices to the original CBMC stages. The purpose of the table is to show which parts of the CBMC flow are extended and which part remains unchanged.

CBMC stage	Component role	Implemented extension	Effect on the checking model
Front-end and goto-cc	Translate C into CBMC’s internal representation.	Read the OSEK/VDX configuration and preserves project context.	Produces the initial checking model used by the later stages.
Interleaving analysis	Not present in the CBMC flow..	Run interleaving-analysis before final verification.	Produces a manifest that records selected ISR insertion locations.
Model building with selected ISR calls	Not present in the CBMC flow..	Use aib with lib-interleaving-block .	Builds a checking model with ISR calls only at the locations recorded in the manifest.
Symbolic execution and goto-symex	Build the solver formula by symbolic execution.	Use the os-api layer to handle recognized OSEK/VDX APIs.	Updates task states, events, the ready queue, and the next running task according to OSEK/VDX scheduling behaviours.
Solver back end	Solve the bounded query and reconstruct counterexamples.	Unchanged.	The final solving step stays the same; the input model has already been refined before solving.

Table 5: Implementation map from CBMC stages to the proposed approach.

4.2 Benchmark Corpus and Experimental Setup

The experimental corpus is drawn from two public benchmark families: i-CBMC [21] and IntAbs [22]. Across the available public measurements, these data reduce to 39 distinct benchmark cases after repeated measurements of the same benchmark family and case name are merged: 18 i-CBMC cases and 21 IntAbs cases.

The experimental setup follows the methodology described in Section 3. For each distinct benchmark case, two versions are checked under the same entry function, checked properties, and unwind bound. The CBMC version represents ISR calls broadly, while the proposed version uses the restricted model-building step.

The main difference is how ISR calls are represented in the checking model. The proposed flow first identifies the ISR entry functions and the program locations that matter for each case. It then builds the checking model with ISR calls only at the selected locations, where they can affect the execution path. Other locations are skipped because they only add unnecessary interleavings.

4.3 Experiments

The evaluation addresses three questions:

1. Does the proposed flow remove spurious bugs caused by unnecessary interleavings in the CBMC version?
2. Does the proposed flow reveal ISR-related behaviours that are not represented by the CBMC version?
3. What time and memory cost is introduced by the additional model-building stages?

To answer these questions, each public case is reported in one of five outcome classes. A case has the *same verification outcome* when both flows end in the same verification class. A case

is in the *CBMC spurious bug removed* class when the CBMC reports failure but the proposed model succeeds and the case audit supports the proposed version as the more accurate checking model. A case is in the *new error exposed by the proposed model* class when the CBMC succeeds but the improved model reaches an error trace that the public audit supports. An *unresolved mismatch* is a case where the two versions disagree, but the available evidence is not sufficient to decide which checking model is more accurate. Finally, a case is *not directly comparable* when the comparison is blocked by a resource limit or when the proposed flow cannot produce usable selected ISR insertion locations.

ID	Comparison outcome	Cases	Interpretation
O1	Same verification outcome	14	Both flows reach the same verification result.
O2	CBMC spurious bug removed	8	The CBMC version reports a failure, but the proposed version succeeds because it removes unnecessary ISR interleavings.
O3	New error exposed by the proposed model	1	The proposed version reaches an error trace that is supported by the public audit, while the CBMC version reports success.
O4	Unresolved mismatch	7	The two versions produce different results, but the available evidence is not sufficient to decide which checking model is more accurate.
O5	Not directly comparable	9	The comparison cannot be made directly because of resource limits or because no usable selected ISR insertion locations are produced.

Table 6: Outcome classes and their IDs across the 39 distinct public benchmark cases.

Table 7 then reports one line per distinct benchmark case and uses the outcome ID from Table 6 to link each case back to its verification class. Repeated public measurements of the same benchmark family and case name are merged before reporting. Peak memory is always taken from the verification phase. For the CBMC version, the time column reports compile-and-verify cost. For the proposed version, the build-model column reports the additional pre-verification stages, while the second time column reports compile-and-verify cost after the restricted checking model has been built.

Table 7: Per-case resource summary keyed by the outcome IDs defined in Table 6. MB reports median verification-phase peak RSS.

Benchmark Case	Outcome ID	CBMC Version		Proposed Version		
		MB	Compile + Verify	MB	Build Model	Compile + Verify
i-CBMC						
blink	O5	14.8	506	20.0	349	1 028
blink-bug	O5	14.7	947	19.7	547	1 590
blink-bug-conc	O5	53.4	2 460	18.4	917	1 662
blink-bug-seq	O5	14 000.0	99	17.3	98	99
blink-conc	O5	56.8	2 483	18.8	1 225	1 473
blink-seq	O5	14 000.2	100	17.3	98	99
brake2-1-bug-po	O4	152.9	3 228	18.0	1 514	1 541
brake2-1-po	O4	148.2	2 579	17.7	2 071	1 409
brake2-2-bug-po	O4	233.6	4 841	18.0	2 152	1 557
brake2-2-po	O4	220.8	10 744	17.7	2 052	1 557
brake2-3-bug-po	O4	291.7	5 202	18.1	1 488	1 597

Table 7 continued from the previous page.

Benchmark Case	Outcome ID	CBMC Version		Proposed Version		
		MB	Compile + Verify	MB	Build Model	Compile + Verify
brake2-3-po	O4	265.3	5 197	17.7	1 993	1 451
logger2-bug-conc	O1	262.2	36 729	16.2	1 314	1 221
logger2-bug-conc-cprover	O2	297.5	55 373	16.3	1 390	890
logger2-conc	O1	262.4	36 278	16.4	1 456	902
logger2-conc-cprover	O1	381.4	90 172	16.2	946	1 030
rc-core	O5	30.6	1 046	32.2	919	2 349
rc-core-bug	O5	30.7	1 098	32.2	880	2 372
IntAbs						
i8xx-tco-1	O2	13.1	1 149	15.3	1 666	1 494
i8xx-tco-2	O2	15.3	987	19.1	1 798	2 141
i8xx-tco-3	O2	15.9	790	19.2	1 799	2 140
i8xx-tco-src-2	O2	15.5	789	19.5	2 313	1 519
i8xx-tco-src-3	O2	17.7	1 062	19.7	2 327	1 527
rmain	O5	30.7	1 223	31.5	1 220	2 184
sc1200wdt-1	O1	12.8	801	15.8	1 759	1 735
sc1200wdt-2	O4	13.0	1 337	15.9	1 985	1 754
sc520wdt-1	O1	14.8	1 176	15.5	1 997	2 150
sc520wdt-2	O3	41.0	1 386	16.7	1 882	1 624
wdt-pci-1	O2	14.5	933	15.8	1 921	2 209
wdt-pci-1-main2	O1	15.3	1 118	18.6	2 766	1 707
wdt-pci-2	O1	10.9	1 200	15.9	2 360	1 976
wdt-pci-2-main2	O1	14.7	333	18.6	860	557
wdt-pci-3	O1	14.5	940	16.6	1 402	2 533
wdt-pci-3-main2	O1	14.7	417	18.6	2 841	2 065
wdt-pci-src-1	O2	16.1	1 492	18.8	2 351	2 194
wdt-pci-src-1-main2	O1	15.2	985	18.6	1 780	1 221
wdt-pci-src-2-main2	O1	14.7	1 108	18.6	2 365	1 832
wdt-pci-src-3	O1	14.2	645	16.2	2 118	2 561
wdt-pci-src-3-main2	O1	15.0	1 257	18.4	2 939	1 123

Tables 6 and 7 show that the proposed approach is mainly useful for building a more accurate checking model. At the same time, by reducing unnecessary interleavings, it can also substantially reduce the checking cost in many cases. The strongest positive result is the group of eight cases classified as O2. In a *logger-oriented* i-CBMC case, the CBMC version reports *VERIFICATION FAILED* with 55 373 ms of compile-and-verify time and 297.5 MB of verification-phase peak RSS. The proposed flow reports *VERIFICATION SUCCESSFUL* for the same bound with 1 390 ms of build-model cost, 890 ms of compile-and-verify time, and 16.3 MB of verification-phase peak RSS. The main point is not only that the proposed version has a lower checking cost in this case. The more important point is that the CBMC flow permits an unnecessary ISR interruption point, which leads to a spurious verification failure. The proposed version checks the ISR entry as one atomic region, so this unnecessary interleaving is removed. As a result, the failure reported by the CBMC version is classified as a spurious bug. The summary also contains one public case in O3, where the proposed model exposes an interrupt-sensitive behavior that is missed by the CBMC version.

The summary also separates cases that should not be counted as clear wins or losses. Seven cases remain unresolved because the available evidence is not sufficient to decide which checking model is more accurate. Nine cases are not directly comparable because the comparison is

blocked by resource limits or by missing selected ISR insertion locations. These cases are reported separately as limitations of the current evaluation, rather than being treated as artificial wins for either version. This makes the experimental summary more conservative and easier to interpret.

The columns in Table 7 should be read as two different views of the checking cost. *MB* reports median verification-phase peak RSS for the checked model. *Compile + Verify* reports the cost of the final checking step. The proposed flow separates *Build Model* from *Compile + Verify* so that the reader can see which cost comes from constructing the restricted checking model and which cost remains in the final verification stage.

The main conclusion from the table is therefore not that the proposed version is uniformly faster. The proposed flow always introduces an upfront *Build Model* cost. However, when the baseline model explores many unnecessary interleavings, the narrower checking model can reduce both verification-phase memory and final compile-and-verify time. When the baseline already has a relatively narrow search space, the reduction during verification may not compensate for the additional model-building cost. The table should therefore be read by considering *Build Model* as the new overhead, *Compile + Verify* as the direct checking cost, and *MB* as an indicator of how much state-space pressure remains during verification.

4.4 Discussion

The main cost of the proposed flow is the additional model-building work performed before final verification. On the 14 distinct benchmark cases where both versions produce the same verification result, the proposed flow is slower in 11 cases. Across this same-result subset, the median end-to-end time increases from 1,113 ms for the CBMC version to 4,001 ms for the proposed version. This result is expected because the proposed flow adds interleaving analysis and model-building stages before the final verification step.

However, this cost is not uniform across all cases. In some benchmark cases, the proposed version performs better than the CBMC flow in terms of checking time or memory usage. This mainly happens when the original program is larger or when the CBMC version explores many unnecessary ISR interleavings. In such cases, the cost of building a more restricted checking model can be compensated by the reduction in behaviours explored during final verification.

For the purpose of this report, this trade-off is reasonable. The objective is not to guarantee faster verification in every case. The first objective is to build a checking model that better reflects OSEK/VDX execution, especially task states, event state, the ready queue, and ISR-related behaviours that can affect the checked execution. The second objective is to make verification more feasible for larger source programs. In such programs, a broad ISR model may create many unnecessary interleavings, which can lead to high memory usage or timeouts in the CBMC flow. By selecting only the ISR insertion locations that can affect the checked execution, the proposed approach can reduce the behaviours explored during final verification. Therefore, the proposed approach mainly aims to make the checking model more accurate. At the same time, it can make verification more feasible for larger source programs where the CBMC flow is limited by high memory usage or timeout caused by unnecessary interleavings.

5 Conclusion

OSEK/VDX applications are important in automotive embedded systems, where software errors can affect reliability and safety. Because these applications consist of multiple tasks dispatched by the OSEK/VDX scheduler, their verification requires a checking model that reflects task states, synchronization events, shared resources, the ready queue, and the task execution order determined by the scheduler. If these behaviours are not represented explicitly, the checking process may explore unnecessary interleavings and may report spurious bugs.

This report presented a checking-model approach for OSEK/VDX applications. The approach

has two main parts. First, it represents the effects of OSEK/VDX APIs on task states, events, shared resources, the ready queue, and task execution order. Second, it uses a manifest to record selected program locations where interrupt service routine (ISR) calls should be inserted. The final checking model contains guarded ISR calls only at these selected locations, instead of inserting ISR calls broadly across the program.

The experimental results show that the proposed approach improves the verification process for OSEK/VDX applications. It builds a more accurate checking model, reduces unnecessary interleavings, and removes spurious bugs reported by the baseline checking process. The results also show an important cost trade-off. The proposed approach adds model-building work before final verification, so it is not guaranteed to be faster in every case. However, in some larger cases, reducing unnecessary ISR interleavings can also reduce time or memory cost and make verification more feasible when the baseline flow is limited by high memory usage or timeout.

Several limitations remain. The supported OSEK/VDX subset can be extended to cover more API behaviours and application patterns. Even with these limitations, the results indicate that constructing a more accurate checking model is a practical direction for verifying OSEK/VDX applications with scheduler-dependent task execution and ISR behaviours.

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